COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 031 - Visualizations and Data Analysis	**SECTION:**	CPE21S1
DATE PERFORMED:	Sep 12, 2026	**DATE SUBMITTED:**	Sep 14, 2026
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ACTIVITY NO. 7 **Purpose Use of Color**

1. PROCEDURE OUTPUT

Task 1

```
import matplotlib.pyplot as plt
import numpy as np
import colorsys


categories = ["A", "B", "C", "D"]
values = [10, 20, 15, 25]


hue_colors = [colorsys.hls_to_rgb(h, 0.5, 0.8) for h in [0.0, 0.25, 0.5, 0.75]]


sat_colors = [colorsys.hls_to_rgb(0.6, 0.5, s) for s in [0.2, 0.4, 0.7, 1.0]]


lum_colors = [colorsys.hls_to_rgb(0.6, l, 0.8) for l in [0.2, 0.4, 0.6, 0.8]]


fig, axes = plt.subplots(1, 3, figsize=(15, 4))

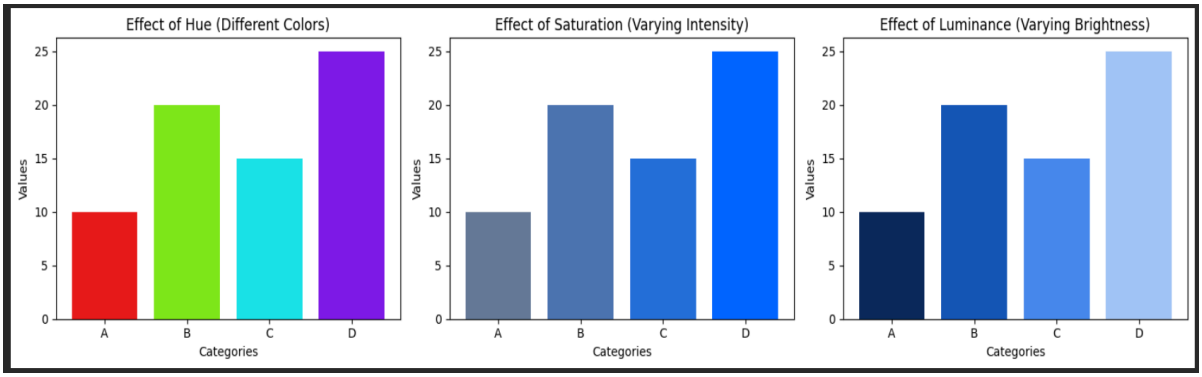

# Chart 1: Varying Hue
axes[0].bar(categories, values, color=hue_colors)
axes[0].set_title("Effect of Hue (Different Colors)")
axes[0].set_xlabel("Categories")
axes[0].set_ylabel("Values")


# Chart 2: Varying Saturation
axes[1].bar(categories, values, color=sat_colors)
axes[1].set_title("Effect of Saturation (Varying Intensity)")
axes[1].set_xlabel("Categories")
axes[1].set_ylabel("Values")


# Chart 3: Varying Luminance
axes[2].bar(categories, values, color=lum_colors)
axes[2].set_title("Effect of Luminance (Varying Brightness)")
axes[2].set_xlabel("Categories")
axes[2].set_ylabel("Values")


plt.tight_layout()
plt.show()
```

Output:


Effect of Hue (Different Colors)
Effect of Saturation (Varying Intensity)
Effect of Luminance (Varying Brightness)
Values
Categories
A
B
C
D
0
5
10
15
20
25

Task 2

```python
import matplotlib.pyplot as plt
import numpy as np

# Sample data for categorical example
categories = ["A", "B", "C", "D", "E"]
values = [10, 30, 20, 40, 25]

# Grid for sequential and diverging examples
x = np.linspace(-3, 3, 100)
y = np.linspace(-3, 3, 100)
X, Y = np.meshgrid(x, y)
Z_seq = np.sqrt(X**2 + Y**2)
Z_div = X

# Create figure
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

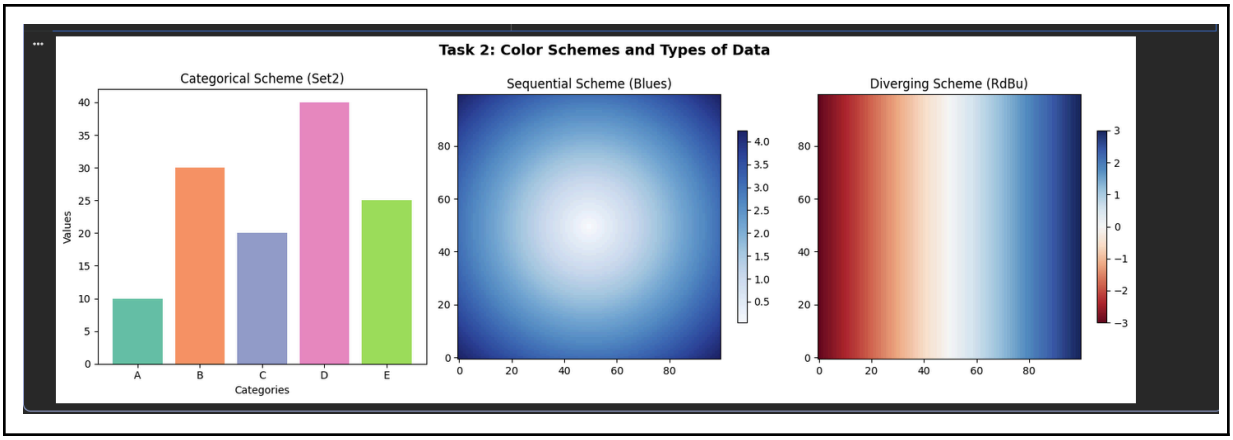
# 1. Categorical Color Scheme
axes[0].bar(categories, values, color=plt.cm.Set2.colors[: len(categories)])
axes[0].set_title("Categorical Scheme")
axes[0].set_xlabel("Categories")
axes[0].set_ylabel("Values")


# 2. Sequential Color Scheme
seq = axes[1].imshow(Z_seq, cmap="Blues", origin="lower")
axes[1].set_title("Sequential Scheme (Blues)")
plt.colorbar(seq, ax=axes[1], shrink=0.7)


# 3. Diverging Color Scheme
div = axes[2].imshow(Z_div, cmap="RdBu", origin="lower", vmin=-3, vmax=3)
axes[2].set_title("Diverging Scheme (RdBu)")
plt.colorbar(div, ax=axes[2], shrink=0.7)


# Layout adjustments
plt.suptitle(
    "Task 2: Color Schemes and Types of Data", fontsize=14, fontweight="bold"
)
plt.tight_layout()
plt.show()
```

Output:


Task 2: Color Schemes and Types of Data
Categorical Scheme (Set2)
Values
Categories
A
B
C
D
E
Sequential Scheme (Blues)
Diverging Scheme (RdBu)

Task 3

```python
import matplotlib.pyplot as plt
import colorsys
import numpy as np

def hsl_to_rgb(h, s, l):
    return colorsys.hls_to_rgb(h, l, s)

num_colors = 5
hues = np.linspace(0, 1, num_colors, endpoint=False)
fixed_saturation = 0.7
fixed_luminance = 0.5

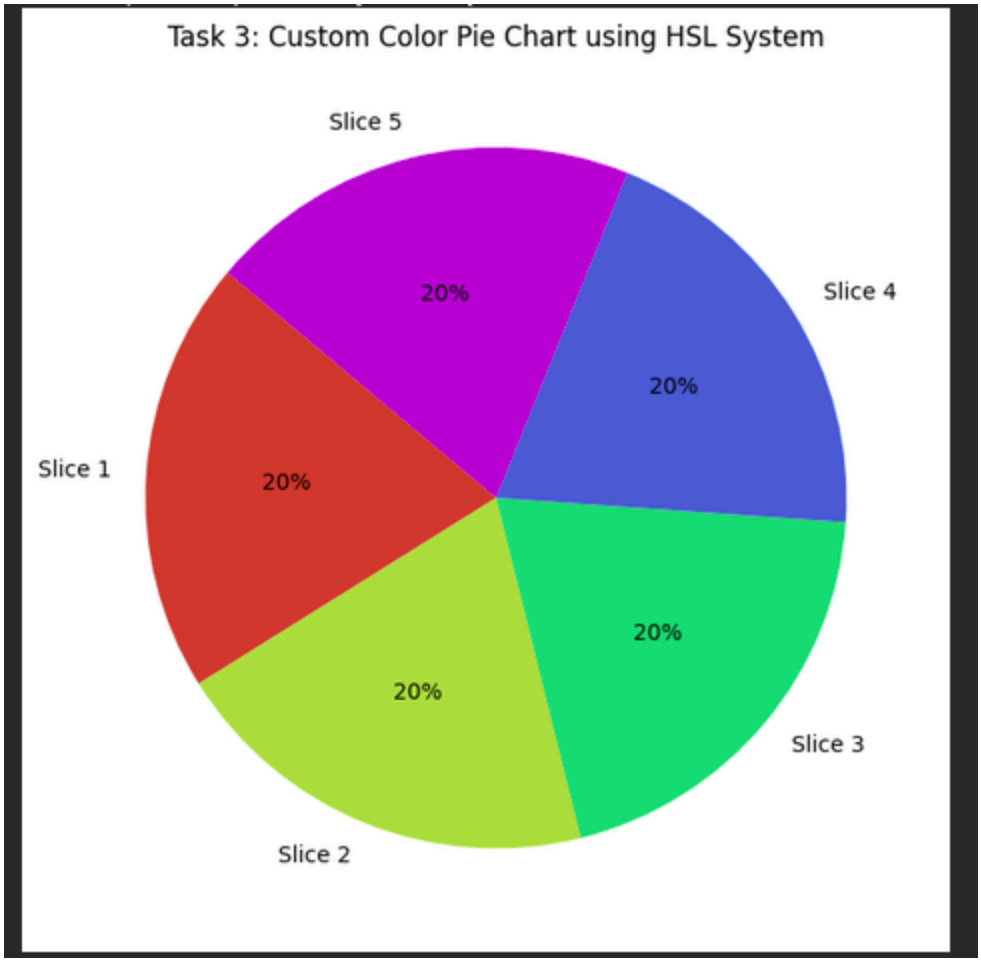

custom_colors = []

for i, h in enumerate(hues):
    rgb = hsl_to_rgb(h, fixed_saturation, fixed_luminance)
    custom_colors.append(rgb)


sizes = [20, 20, 20, 20, 20]
labels = [f"Slice {i+1}" for i in range(num_colors)]

plt.figure(figsize=(6, 6))
plt.pie(
    sizes,
    labels=labels,
    colors=custom_colors,
    autopct="%1.0f%%",
    startangle=140,
)
plt.title("Task 3: Custom Color Pie Chart using HSL System")
plt.tight_layout()
plt.show()
```

Outpu:t


Task 3: Custom Color Pie Chart using HSL System
Slice 5
20%
Slice 4
20%
Slice 1
20%
20%
20%
Slice 3
Slice 2

Task 4

```python
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Sample data
categories = ["A", "B", "C", "D"]
values = [10, 25, 17, 30]

fig, axs = plt.subplots(3, 2, figsize=(12, 10))

# 1. Unnecessary Color (Bad)
axs[0,0].bar(categories, values, color=["red", "blue", "green", "purple"])
axs[0,0].set_title("Bad: Unnecessary Colors")

# Good: One meaningful color
axs[0,1].bar(categories, values, color="steelblue")
axs[0,1].set_title("Good: Simple Color")

# 2. Excessive Color (Bad)
colors = sns.color_palette("hsv", 12)
axs[1,0].bar(range(12), np.random.randint(5,20,12), color=colors)
axs[1,0].set_title("Bad: Excessive Colors")

# Good: Limited palette
sns.barplot(x=categories, y=values, hue=categories, palette="Set2", legend=False, ax=axs[1,1])
axs[1,1].set_title("Good: Limited Colors")

# 3. Insufficient Contrast (Bad)
axs[2,0].bar(categories, values, color="yellow")
axs[2,0].set_facecolor("white")
axs[2,0].set_title("Bad: Low Contrast")

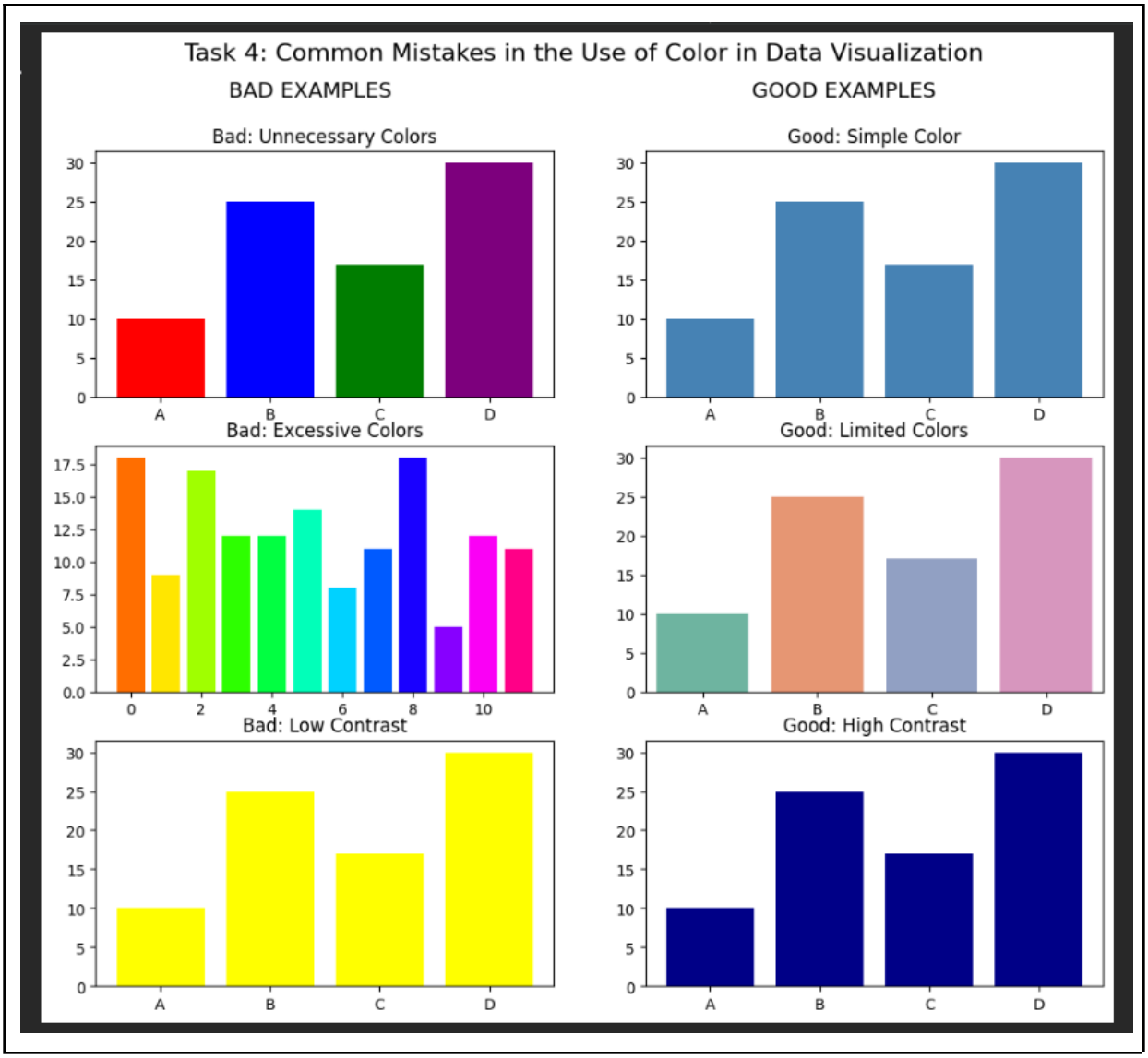
# Good: High Contrast
axs[2,1].bar(categories, values, color="darkblue")
axs[2,1].set_facecolor("white")
axs[2,1].set_title("Good: High Contrast")

#Title
fig.suptitle(
    "Task 4: Common Mistakes in the Use of Color in Data Visualization",
    fontsize=16)

# Labels
fig.text(0.29, 0.93, "BAD EXAMPLES", ha="center", fontsize=14)
fig.text(0.70, 0.93, "GOOD EXAMPLES", ha="center", fontsize=14)

plt.tight_layout
plt.show()
```

Output:


Task 4: Common Mistakes in the Use of Color in Data Visualization
BAD EXAMPLES
GOOD EXAMPLES
Bad: Unnecessary Colors
Good: Simple Color
Bad: Excessive Colors
Good: Limited Colors
Bad: Low Contrast
Good: High Contrast
A
B
C
D

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Code

```
import matplotlib.pyplot as plt
import matplotlib.colors as mcolors
import seaborn as sns
import colorsys
import numpy as np
import pandas as pd

# Step 1: Create dataset
df = pd.read_csv("/content/student_performance_dataset.csv")
df["internet_access_num"] = df["internet_access"].map({"Yes": 1, "No": 0})
df["part_time_job_num"] = df["part_time_job"].map({"Yes": 1, "No": 0})

categories = ["study_time_hours", "attendance_percent", "sleep_hours",
              "internet_access", "part_time_job"]
raw_cols = ["study_time_hours", "attendance_percent", "sleep_hours",
            "internet_access_num", "part_time_job_num"]
values = [round(df[c].corr(df["final_exam_score"]), 4) for c in raw_cols]
average = np.mean(values)
n = len(categories)

# ==============================
# Step 2: Color and Perception
# ==============================

# Hue
plt.bar(categories, values, color=["red", "blue", "green", "orange", "purple"])
plt.title("Hue Example")
plt.ylabel("Correlation with Final Exam Score")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()
```

```python
# TODO: Add Saturation Example
fixed_hue = 0.6  # blue
saturations = [0.1, 0.3, 0.5, 0.7, 0.9]
sat_colors = [colorsys.hls_to_rgb(fixed_hue, 0.5, s) for s in saturations]
plt.bar(categories, values, color=sat_colors)
plt.title("Saturation Example")
plt.ylabel("Correlation with Final Exam Score")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()

# TODO: Add Luminance Example
lightnesses = [0.2, 0.35, 0.5, 0.65, 0.8]
lum_colors = [colorsys.hls_to_rgb(fixed_hue, l, 0.6) for l in lightnesses]
plt.bar(categories, values, color=lum_colors)
plt.title("Luminance Example")
plt.ylabel("Correlation with Final Exam Score")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()

# ==============================
# Step 3: Color Schemes
# ==============================

# Categorical
sns.barplot(x=categories, y=values, hue=categories, palette="Set2", legend=False)
plt.title("Categorical Scheme")
plt.ylabel("Correlation with Final Exam Score")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()
```

```python
# TODO: Add Sequential Scheme
seq_norm = mcolors.Normalize(vmin=min(values), vmax=max(values))
seq_cmap = plt.get_cmap("Blues")
seq_colors = [seq_cmap(seq_norm(v)) for v in values]
plt.bar(categories, values, color=seq_colors)
plt.title("Sequential Scheme")
plt.ylabel("Correlation with Final Exam Score")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()

# TODO: Add Diverging Scheme
div_norm = mcolors.TwoSlopeNorm(vmin=min(values), vcenter=0, vmax=max(values))
div_cmap = plt.get_cmap("RdBu")  # red = negative relationship, blue = positive
div_colors = [div_cmap(div_norm(v)) for v in values]
plt.bar(categories, values, color=div_colors)
plt.axhline(0, color="black", linestyle="--", linewidth=1, label="No correlation (0)")
plt.title("Diverging Scheme")
plt.ylabel("Correlation with Final Exam Score")
plt.xticks(rotation=20, ha="right")
plt.legend()
plt.tight_layout()
plt.show()


# ============================
# Step 4: Custom Colors with HSL
# ============================

# Generate HSL -> RGB
hsl_colors = [(i / n, 0.6, 0.5) for i in range(n)]
rgb_colors = [colorsys.hls_to_rgb(h, l, s) for h, l, s in hsl_colors]
```

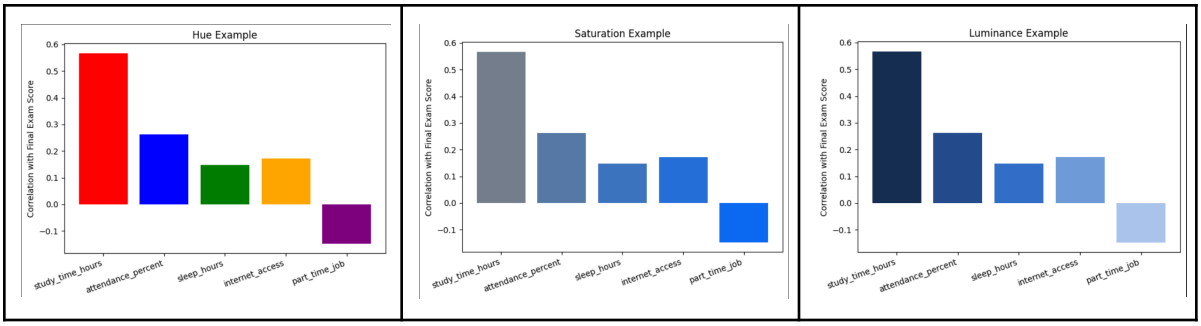
```python
# Pie chart
plt.pie([abs(v) for v in values], labels=categories, colors=rgb_colors, autopct="%1.1f%%")
plt.title("Custom HSL Colors")
plt.tight_layout()
plt.show()

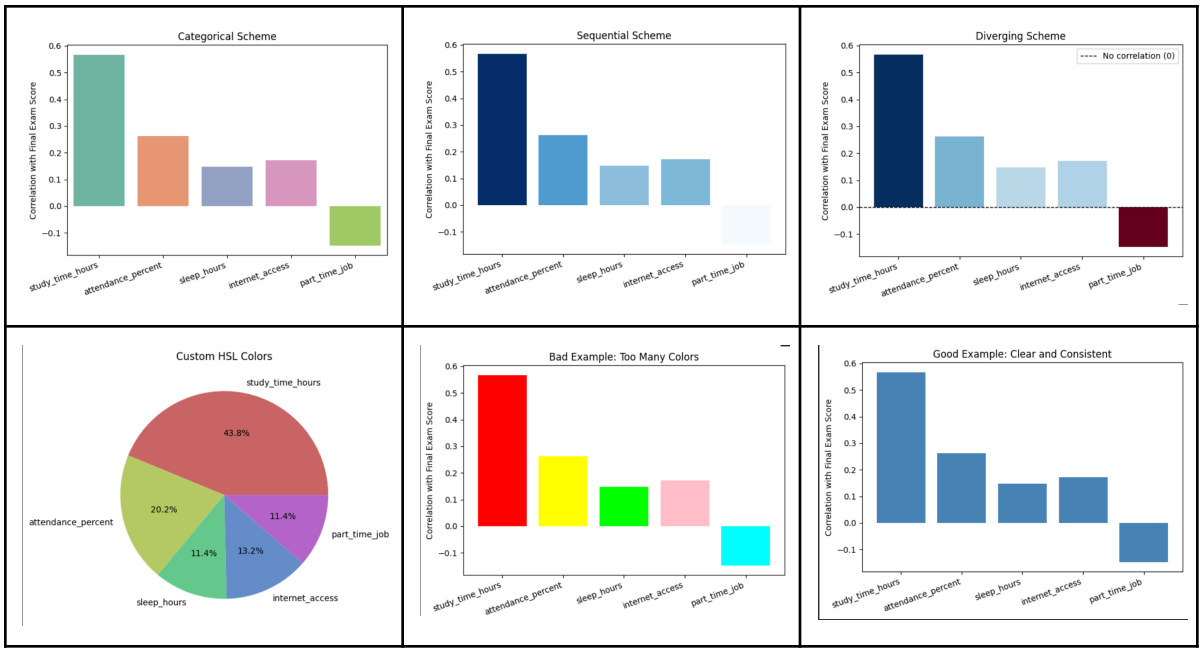

# ==============================
# Step 5: Common Mistakes
# ==============================

# Bad Example
plt.bar(categories, values, color=["red", "yellow", "lime", "pink", "cyan"])
plt.title("Bad Example: Too Many Colors")
plt.ylabel("Correlation with Final Exam Score")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()

# Good Example
plt.bar(categories, values, color="steelblue")
plt.title("Good Example: Clear and Consistent")
plt.ylabel("Correlation with Final Exam Score")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()
```

Output:





Task Description:
This project is about learning to use color on purpose in data visualizations, not just for decoration, but to actually help someone read the data correctly. It uses a real dataset (student performance factors vs. final exam scores) to walk through the main ideas behind color theory in charts, in five parts.

Dataset:
Five student factors (study_time_hours, attendance_percent, sleep_hours, internet_access, part_time_job) are each reduced to one number: their correlation with final_exam_score.

Problem Statement:
Academic performance is shaped by more than just intelligence, daily habits, resources, and commitments outside the classroom all play a role. This task investigates the relationship between five student-level factors (study_time_hours, attendance_percent, sleep_hours, internet_access, and part_time_job) and final_exam_score, using purposeful color choices to make the strength and direction of each relationship immediately readable.

Conclusion

Canoy	In this activity I learned a lot. It was complicated at first but I got the hang of it. I learned how the concept of purposeful use of color works and how it can make data easier to understand. I also learned how hue, saturation, luminance, and different color schemes can be used in charts and how to use them and what's good and bad for the charts. Overall, I learned more on how to apply colors in code and choose colors that make the visualization clear instead of just using them for decoration.
Corro	This activity taught me the importance of choosing the right colors for graphs and charts. I learned how to use contrast, custom color schemes, and HSL settings to make the data clear and easy to read. I also learned how to make charts accessible to colorblind readers and how to keep the design clean. Overall this exercise was helpful in making me understand how to build better and clearer visualizations.
De Leon	Out of the five factors I looked at, I found that study_time_hours had the strongest link to final exam scores (r = 0.57), much stronger than anything else I tested. I also saw that attendance_percent (r = 0.26), internet_access (r = 0.17), and sleep_hours (r = 0.15) helped, but only a little. Also, color should always represent something in the data, not just decorate it — the type of data (categorical, ordered, or diverging) should decide the color scheme. Random colors are confusing; purposeful color makes data clearer and easier to trust.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT and Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction, Guide, Ideas and Error Checking.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Used it only as a guide for grammar, brainstorming ideas..**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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